

PORTFOLIO

오 승 훈

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PRESONAL INFORMATION

01



오승훈

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EDUCATION

2019.03~2025.02

한림대학교 빅데이터학과 졸업

EXPERIENCE

2022.03~2025.04

AIAC LAB (AI ACCELERATOR COMPUTING LAB)

2025.01~2025.03

ELECTRONICS AND TELECOMMUNICATIONS
RESEARCH INSTITUTE (INTERN)

COMPETITION

2022

2022 국방 AI 경진대회 - 특별상(4등)

2023

2023 SW 중심대학 공동 AI 경진대회 - SW중심대학협의회장상

2024

2024 SW 중심대학 공동 AI 경진대회 - 특별상(4등)

2024

제3회 ETRI 휴먼이해 인공지능 경진대회 - 과학기술정보통신부장관상

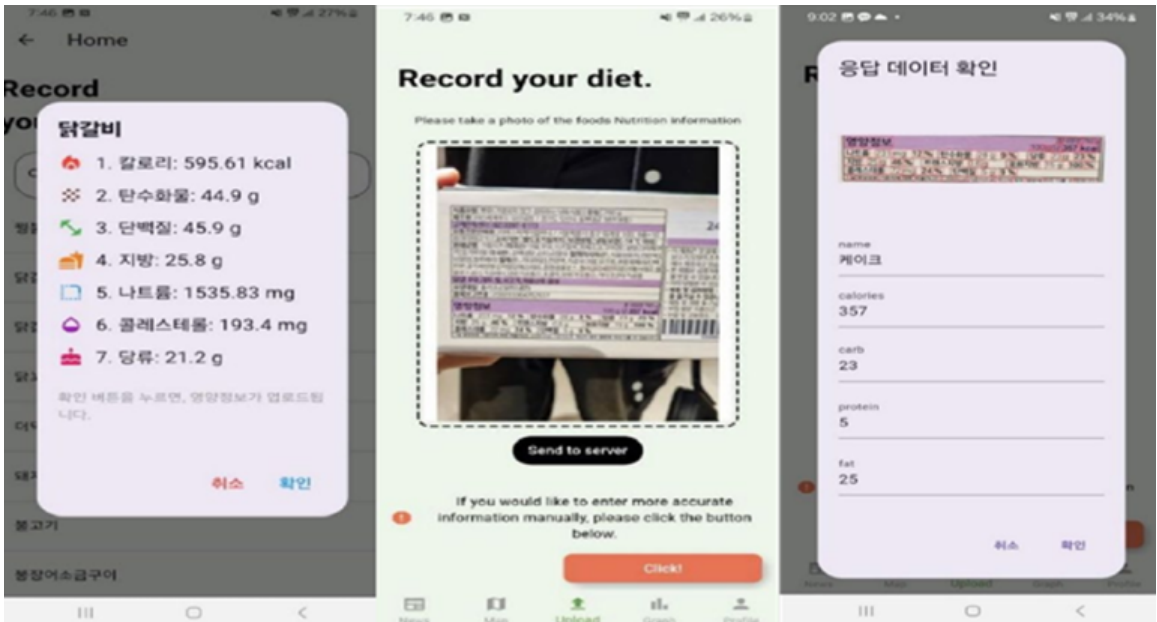
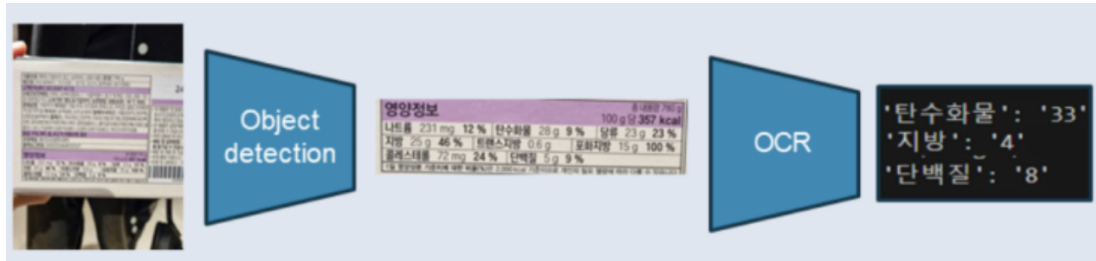
PROJECT

CAPSTONE DESIGN

COWEKRER - 학부 4명

DATE - 2024.02 ~ 2024.06

SKILL - OBJECT DETECTION, OCR



PURPOSE

현대 사회에서는 배달 음식과 가공식품의 발달로 비만율과 질병이 많이 발생하고 있다. 이러한 점을 방지하기 위해 사용자에게 맞는 영양섭취 정보를 보다 편리하게 기입하여 정리 할 수 있도록 해주는 어플리케이션을 만들기 위함

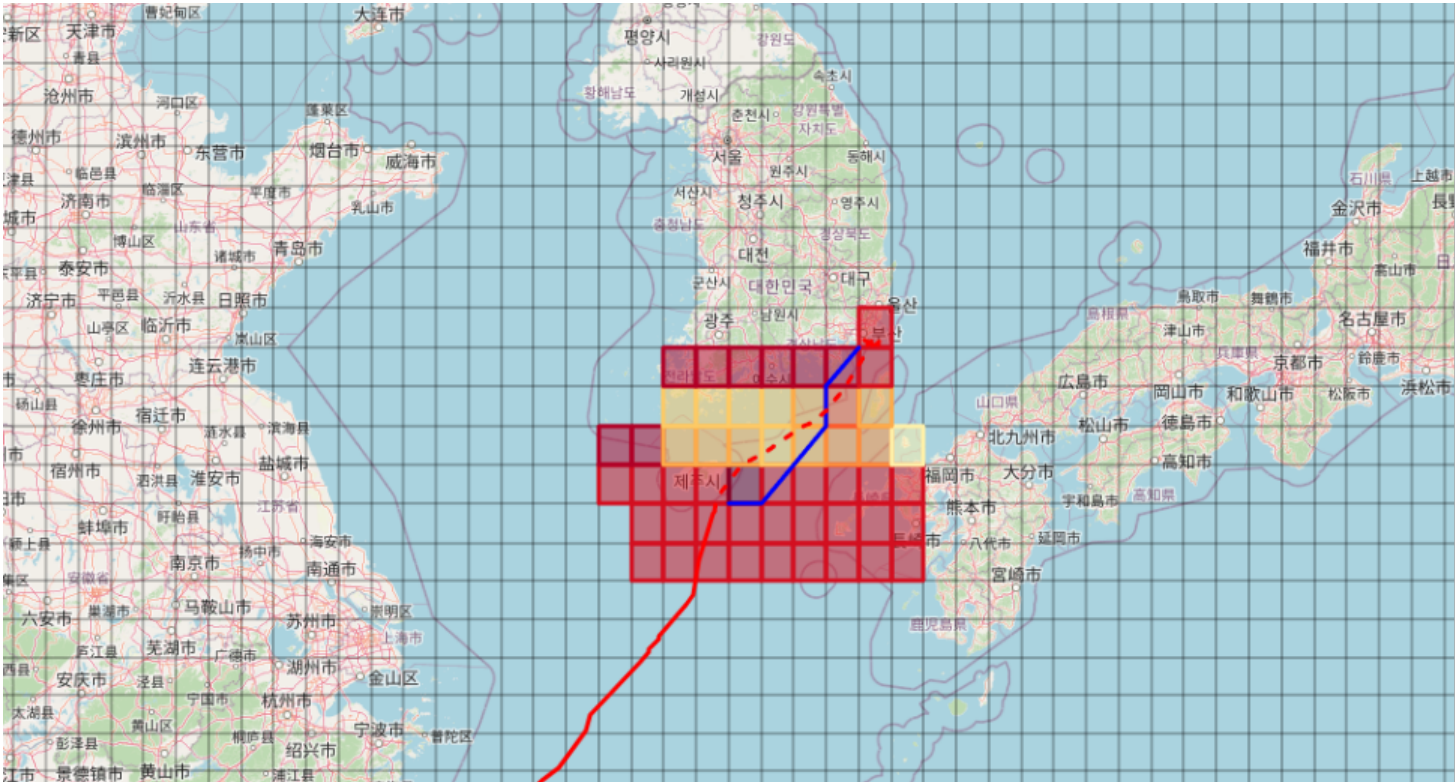
PROJECT

SEAGUARD

COWEKRER - 학부 4명

DATE - 2023.08

SKILL - MACHINE LEARNING



PURPOSE

선박 운항에 있어서 사고를 예방하고자 해양의 상태에 따른 선박의 경로
조정 프로그램 생성

PROJECT

BOUNDARY UNLEARNING IMPLEMENTATION

COWEKRER - X

DATE - 2024.10

SKILL - BOUNDARY UNLEARNING



PURPOSE

MACHINE UNLEARNING, 특정 데이터를 학습한 모델에서 해당 데이터를 기반으로 한 학습 기록을 지우는 것.
꾸준히 증가하고 있는 DEEPFAKE 이슈에 따른 관련된 논문으로, 학습된 DNN 모델에서 BOUNDARY DECISION을 이동시켜 전체 클래스를 잊게 만드는 연구

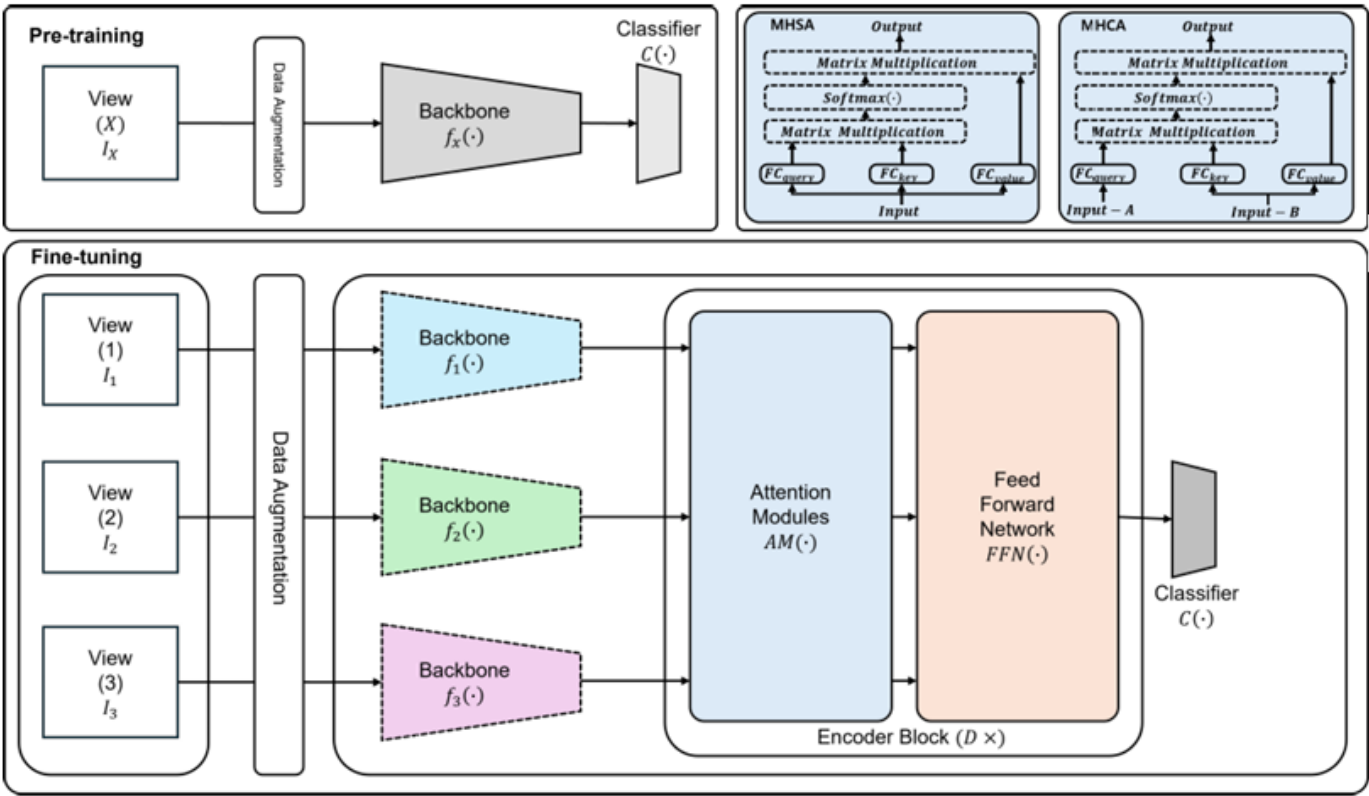
PROJECT

AHI CLASSIFICATION USING PATIENT'S FACE IMAGE

COWEKRER - 2명

DATE - 2024.10 ~ 2025.01

SKILL - CROSS ATTENTION



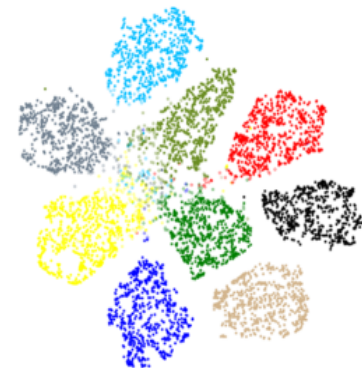
PURPOSE

환자의 얼굴 이미지를 활용한 수면무호흡증 중증도 예측
(TRANSFORMER 기반 AHI 분류)

PAPER REVIEW

BOUNDARY UNLEARNING : RAPID FORGETTING OF DEEP NETWORKS VIA SHIFTING THE DECISION BOUNDARY

Visualization of decision space

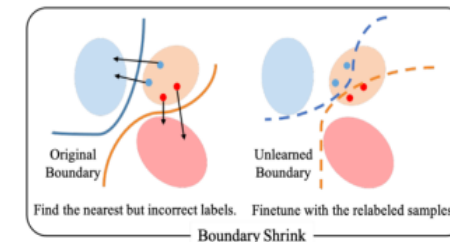


About Machine Unlearning

- Retrain is the simplest method.
→ Time consuming and expensive
- The decision boundary of the sample to forget is broken that the samples to forget are spread around the decision space of the retrain DNN model.
- Most samples to forget move to the boundary of other clusters, which means that samples located at the boundary of the cluster are likely to have great uncertainty in prediction.



Boundary Unlearning – Boundary Shrink



$$\mathbf{w}_0 = \arg \min_{\mathbf{w}} \sum_{(\mathbf{x}_i, \mathbf{y}_i) \in \mathcal{D}} \mathcal{L}(\mathbf{x}_i, \mathbf{y}_i, \mathbf{w})$$

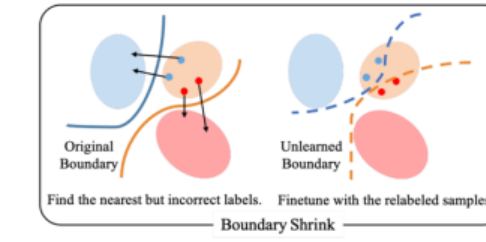
The above expression is a definition of parameter $\mathbf{w}_0$, which is the optimal solution for the original model. For each dataset in the training dataset, the loss function is defined and updated in the direction that the value is minimized.

$$\mathbf{x}'_f = \mathbf{x}_f + \epsilon \cdot \text{sign}(\nabla_{\mathbf{x}_f} \mathcal{L}(\mathbf{x}_f, \mathbf{y}, \mathbf{w}_0))$$

- Among the training datasets, $\mathbf{x}_f$ data, adds noise as shown above.
- Find the slope of the input data $\mathbf{x}_f$ for the loss function of the original model, which gives gradient information about how the original model perceives the forgetting data.
- This is a process to extract only the directionality of the gradients obtained in the above process, leaving only information about whether each direction is positive or negative.
- Add a constant magnitude of noise magnitude to the directionality of the gradients obtained above.



Boundary Unlearning – Boundary Shrink



$$\mathbf{y}_{nbi} \leftarrow \text{softmax}(f_{\mathbf{w}_0}(\mathbf{x}'_f))$$

Enter $\mathbf{x}'_f$ the noise-added forgetting data, into the original model to extract the softmax value, which gives $\mathbf{y}_{nbi}$, another closet label to the existing label $\mathbf{y}$.

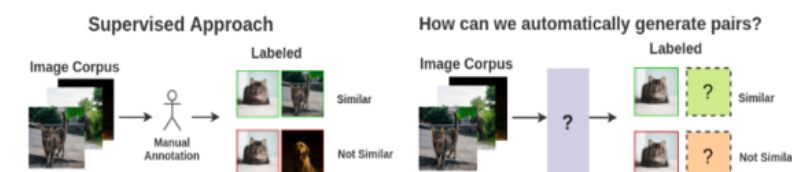
$$\mathbf{w}' = \arg \min_{\mathbf{w}} \sum_{(\mathbf{x}_i, \mathbf{y}_{nbi}) \in \mathcal{D}_f} \mathcal{L}(\mathbf{x}_i, \mathbf{y}_{nbi}, \mathbf{w})$$

- Finally, $\mathbf{y}_{nbi}$ labels are assigned to $\mathbf{x}_f$ to finetune, moving the data to the most similar class. This allows the model to adjust the decision boundaries while preserving the maximum relationship between the learned data.

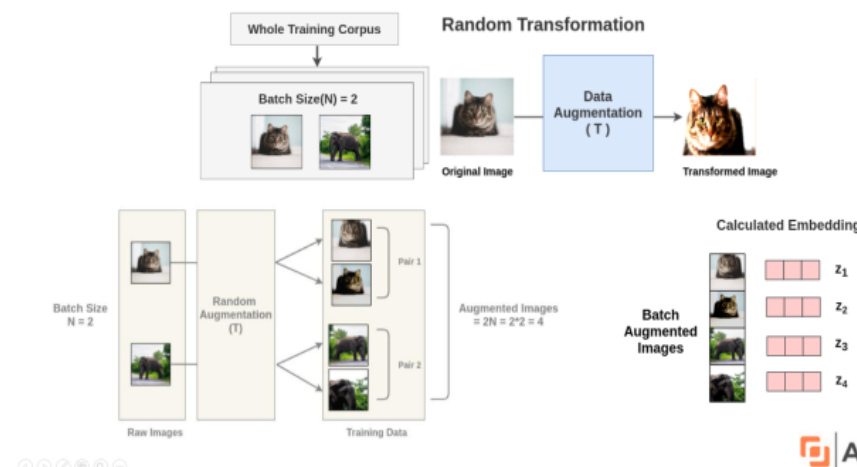


SIMCLR : A SIMPLE FRAMEWORK FOR CONTRASTIVE LEARNING OF VISUAL REPRESENTATIONS

Contrastive Learning



SimCLR Framework



SimCLR Loss

$$\text{similarity}(\mathbf{x}_i, \mathbf{x}_j) = \text{cosine similarity}(\mathbf{z}_i, \mathbf{z}_j)$$



$$\ell_{i,j} = -\log \frac{\exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_j)/\tau)}{\sum_{k=1}^{2N} \mathbb{1}_{[k \neq i]} \exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_k)/\tau)}$$

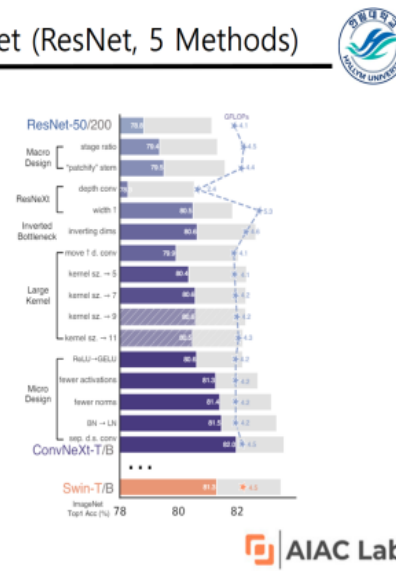


PAPER REVIEW

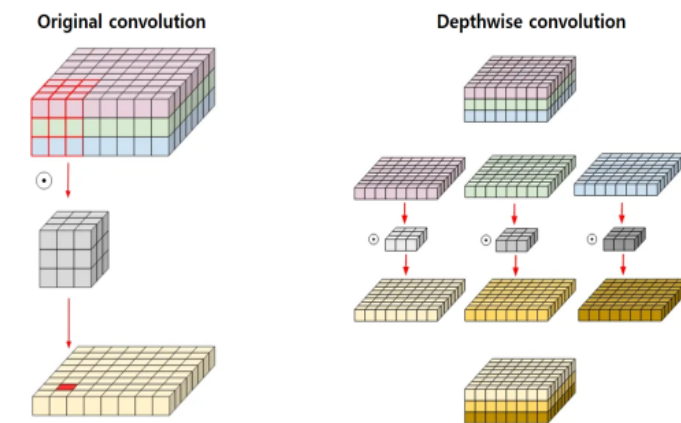
CONVNET : A CONVNET FOR THE 2020S

How to modernize a standard ConvNet (ResNet, 5 Methods)

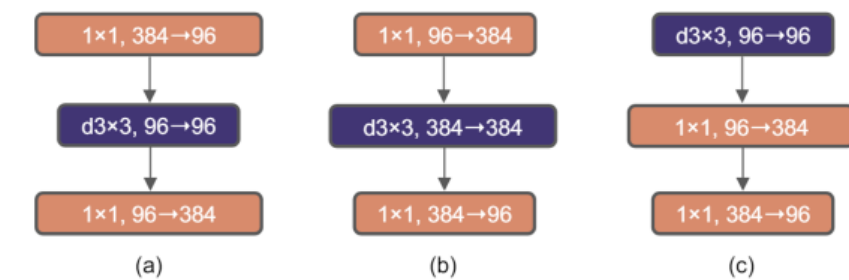
1. Macro Design
2. ResNeXt-ify
3. Inverted Bottleneck
4. Large Kernel Size
5. Micro Design



ResNeXt-ify (Depthwise convolution)

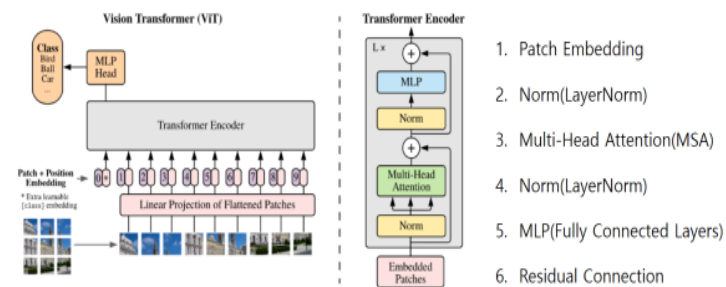


Inverted Bottleneck



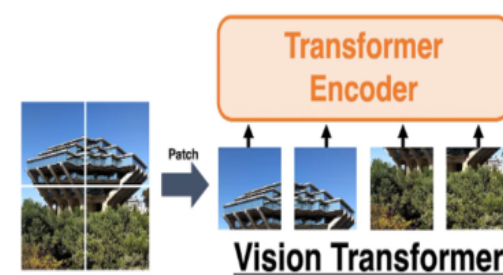
VIT : AN IMAGE IS WORTH 16X16 WORDS: TRANSFORMERS FOR IMAGE RECOGNITION AT SCALE

Vision Transformer Architecture

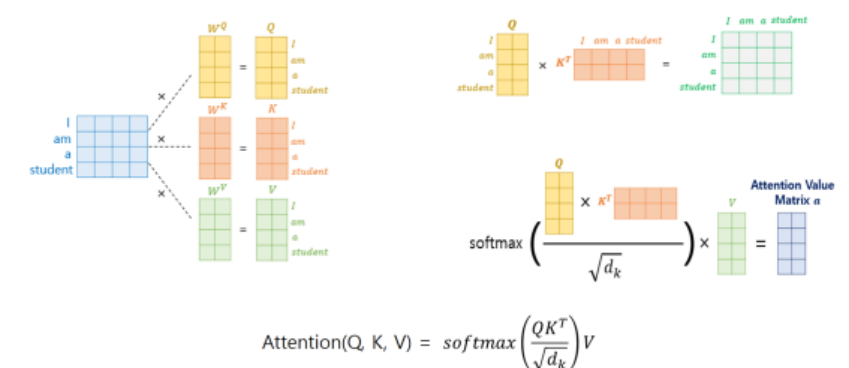


1. Patch Embedding
2. Norm(LayerNorm)
3. Multi-Head Attention(MSA)
4. Norm(LayerNorm)
5. MLP(Fully Connected Layers)
6. Residual Connection

Image Patch



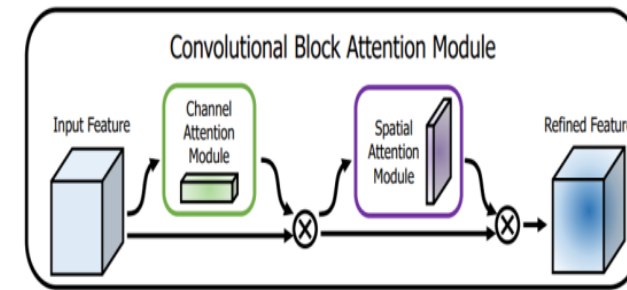
Self Attention



PAPER REVIEW

CBAM : CONVOLUTION BLOCK ATTENTION MODULE

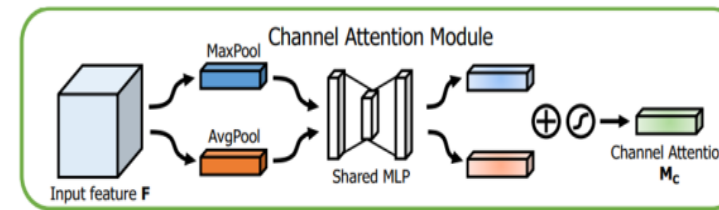
Convolution Block Attention Module



$$\begin{aligned} F' &= M_c(F) \otimes F, \\ F'' &= M_s(F') \otimes F', \end{aligned}$$

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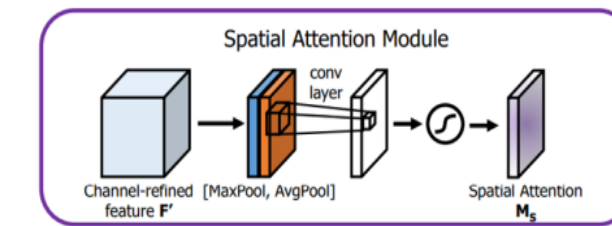
Channel Attention Module



$$\begin{aligned} M_c(F) &= \sigma(MLP(AvgPool(F)) + MLP(MaxPool(F))) \\ &= \sigma(W_1(W_0(F_{avg}^c) + W_0(F_{max}^c))), \end{aligned}$$

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Spatial Attention Module



$$\begin{aligned} M_s(F) &= \sigma(f^{7 \times 7}([AvgPool(F); MaxPool(F)])) \\ &= \sigma(f^{7 \times 7}([F_{avg}^s; F_{max}^s])), \end{aligned}$$

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VIT : AN IMAGE IS WORTH 16X16 WORDS: TRANSFORMERS FOR IMAGE RECOGNITION AT SCALE

Main Idea – GradCAM

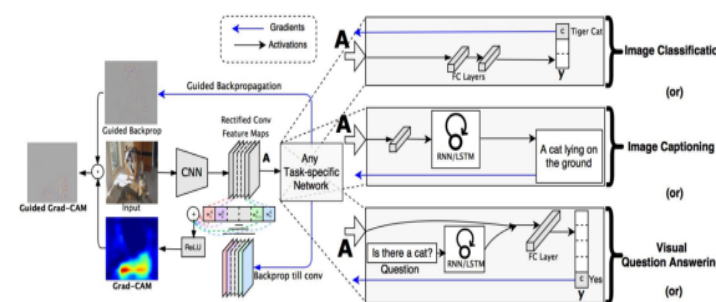


Fig. 2: Grad-CAM overview: Given an image and a class of interest (e.g., "tiger cat" or any other type of differentiable output) as input, we forward propagate the image through the CNN part of the model and then through task-specific computations to obtain a raw score for the category. The gradients are set to zero for all classes except the desired class (tiger cat), which is set to 1. This signal is then backpropagated to the rectified convolutional feature maps of interest, which we combine to compute the coarse Grad-CAM localization (blue heatmap) which represents where the model has to look to make the particular decision. Finally, we pointwise multiply the heatmap with guided backpropagation to get Guided Grad-CAM visualizations which are both high-resolution and concept-specific.

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Equation - CAM

$$\begin{aligned} \text{Class score} \\ S_c &= \sum_k w_k^c \sum_{x,y} f_k(x,y) \\ &= \sum_{x,y} \sum_k w_k^c f_k(x,y). \\ S_c &= \sum_{x,y} M_c(x,y) \end{aligned}$$

Class Activation Map for class c

$$M_c(x,y) = \sum_k w_k^c f_k(x,y).$$

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Counterfactual Explanations

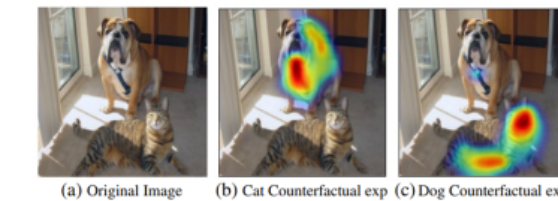


Fig. 3: Counterfactual Explanations with Grad-CAM

$$\begin{aligned} \text{Global average pooling} \\ \alpha_k^c &= \frac{1}{Z} \sum_i \sum_j - \frac{\partial y^c}{\partial A_{ij}^k} \\ \text{Negative gradients} \end{aligned}$$

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 **HANK YOU**
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